# MTH 201 Chapter 2: Integration Techniques

## 2.7 Improper Integrals Worksheet

1. Evaluate $\int_1^{\infty} e^{-x}dx$ if possible.

$$\int_1^{\infty} e^{-x}dx = \lim_{t\to\infty}\left[\int_1^t e^{-x}\,dx\right] = \lim_{t\to\infty}\left[-e^{-x}\Big|_1^t\right] = \lim_{t\to\infty}\left[-\frac{1}{e^t}+\frac{1}{e}\right] = 0+\frac{1}{e} = \frac{1}{e}$$

The integral converges to $\dfrac{1}{e}$

2. Evaluate $\int_{-8}^{1} \frac{1}{\sqrt[3]{x}}dx$ if possible.

There is a discontinuity at 0. So, we write:

$$\int_{-8}^{1} \frac{1}{\sqrt[3]{x}}dx = \int_{-8}^{0} \frac{1}{\sqrt[3]{x}}dx + \int_{0}^{1} \frac{1}{\sqrt[3]{x}}dx = \lim_{t\to 0^-}\int_{-8}^{t} \frac{1}{\sqrt[3]{x}}dx + \lim_{s\to 0^+}\int_{s}^{1} \frac{1}{\sqrt[3]{x}}dx$$

$$\lim_{t\to 0^-}\left(\frac{3}{2}x^{\frac{2}{3}}\Big|_{-8}^{t}\right) + \lim_{s\to 0^+}\left(\frac{3}{2}x^{\frac{2}{3}}\Big|_{s}^{1}\right) = \lim_{t\to 0^-}\left(\frac{3}{2}t^{\frac{2}{3}}-6\right) + \lim_{s\to 0^+}\left(\frac{3}{2}-\frac{3}{2}s^{\frac{2}{3}}\right) = \frac{-9}{2}$$

3. Evaluate $\int_0^1 \ln(x)dx$ if possible.

There is a discontinuity at 0. So, we write: $\int_0^1 \ln(x)dx = \lim_{t\to 0^+}\int_t^1 \ln(x)dx$. We can use integration by parts to evaluate the integral on the right. In particular we let $u=\ln(x)$ and $dv = dx$. These two equations imply $du = \frac{1}{x}dx$ and $v = x$. Now, we apply the integration by parts formula and we obtain:

$$\lim_{t\to 0^+}\int_t^1 \ln(x)dx = \lim_{t\to 0^+}\left(x\ln(x)\Big|_t^1 - \int_t^1 dx\right) = \lim_{t\to 0^+}\left[x\ln(x)\Big|_t^1 - x\Big|_t^1\right]$$

$$= \lim_{t\to 0^+}\left[\left(\ln(1)-t\ln(t)-(1-t)\right)\right] = \lim_{t\to 0^+}\left[-t\ln(t)-1+t\right]$$

$$= \lim_{t\to 0^+}\left(\frac{-\ln(t)}{1/t}\right)-1 \overset{H}{=} \lim_{t\to 0^+}\left(\frac{-1/t}{-1/t^2}\right)-1 = \lim_{t\to 0^+}\left(\frac{1}{1/t}\right)-1 = \lim_{t\to 0^+}(t)-1 = -1$$

4. Evaluate $\int_{-\infty}^{3} \frac{e^x}{3-2e^x}\,dx$ if possible.

We note that $\int_{-\infty}^{3} \frac{e^x}{3-2e^x}\,dx = \lim_{t\to-\infty}\left[\int_{t}^{3} \frac{e^x}{3-2e^x}\,dx\right]$.

Let $u = 3-2e^x$ and then $du = -2e^x dx \;\to\; -\frac{1}{2}du = e^x dx$. The bounds become $u(t) = 3-2e^t$ & $u(3) = 3-2e^3$. Therefore,

$$\lim_{t\to-\infty}\left[\int_{t}^{3} \frac{e^x}{3-2e^x}\,dx\right] = \lim_{t\to-\infty}\left[\int_{3-2e^t}^{3-2e^3} -\frac{1}{2}\cdot\frac{1}{u}\,du\right] = \lim_{t\to-\infty}\left[-\frac{1}{2}\ln|u|\Big|_{3-2e^t}^{3-2e^3}\right] = \lim_{t\to-\infty}\left[-\frac{1}{2}\ln\left|3-2e^3\right| + \frac{1}{2}\ln\left|3-2e^t\right|\right]$$

$$= -\frac{1}{2}\ln\left|3-2e^3\right| + \frac{1}{2}\ln(3)$$

5. Evaluate $\int_{-\infty}^{\infty} \frac{dx}{\sqrt[3]{x-1}}$ if possible.

We first note that $\int_{-\infty}^{\infty} \frac{dx}{\sqrt[3]{x-1}} = \int_{-\infty}^{0} \frac{dx}{\sqrt[3]{x-1}} + \int_{0}^{\infty} \frac{dx}{\sqrt[3]{x-1}} = \lim_{t\to-\infty}\left[\int_{t}^{0} \frac{dx}{\sqrt[3]{x-1}}\right] + \lim_{s\to\infty}\left[\int_{0}^{s} \frac{dx}{\sqrt[3]{x-1}}\right]$

We calculate that

$$\lim_{t\to-\infty}\left[\int_{t}^{0} \frac{dx}{\sqrt[3]{x-1}}\right] = \lim_{t\to-\infty}\left[\frac{3}{2}(x-1)^{2/3}\Big|_{t}^{0}\right] = \lim_{t\to-\infty}\left[\frac{3}{2}(-1)^{2/3} - \frac{3}{2}(t-1)^{2/3}\right] = \frac{3}{2} - \frac{3}{2}\lim_{t\to-\infty}(t-1)^{2/3} = -\infty$$

Since the first integral here diverges we know that $\int_{-\infty}^{\infty} \frac{dx}{\sqrt[3]{x-1}}$ diverges overall.

6. Evaluate $\int_{-1}^{\infty} \frac{x^2}{\sqrt{x^3+1}}\,dx$ if possible.

We first note that

$$\int_{-1}^{\infty} \frac{x^2}{\sqrt{x^3+1}}\,dx = \int_{-1}^{0} \frac{x^2}{\sqrt{x^3+1}}\,dx + \int_{0}^{\infty} \frac{x^2}{\sqrt{x^3+1}}\,dx$$

$$= \lim_{t\to -1^+}\left[\int_{t}^{0} \frac{x^2}{\sqrt{x^3+1}}\,dx\right] + \lim_{s\to\infty}\left[\int_{0}^{s} \frac{x^2}{\sqrt{x^3+1}}\,dx\right]$$

We calculate that

$$\lim_{t\to -1^+}\left[\int_{t}^{0} \frac{x^2}{\sqrt{x^3+1}}\,dx\right] = \lim_{t\to -1^+}\left[\int_{t^3+1}^{1} \frac{1}{3}\frac{1}{\sqrt{u}}\,du\right] = \lim_{t\to -1^+}\left[\frac{2}{3}u^{1/2}\Big|_{t^3+1}^{1}\right] = \lim_{t\to -1^+}\left[\frac{2}{3}\sqrt{1} - \frac{2}{3}\sqrt{t^3+1}\right] = \frac{2}{3}$$

$$\lim_{s\to\infty}\left[\int_{0}^{s} \frac{x^2}{\sqrt{x^3+1}}\,dx\right] = \lim_{s\to\infty}\left[\int_{1}^{s^3+1} \frac{1}{3}\frac{1}{\sqrt{u}}\,du\right] = \lim_{s\to\infty}\left[\frac{2}{3}\sqrt{u}\Big|_{1}^{s^3+1}\right] = \lim_{s\to\infty}\left[\frac{2}{3}\sqrt{s^3+1} - \frac{2}{3}\right] = \infty - \frac{2}{3} = \infty$$

Since one part of the overall integral diverges, the entire integral diverges.

7. Use the Comparison Theorem to determine whether the integral is convergent or divergent.

a) $\displaystyle\int_1^\infty \frac{x^4}{x^5-3x}\,dx$

For $x \geq 1$ we see that $\dfrac{x^4}{x^5-3x} > \dfrac{x^4}{x^5} = \dfrac{1}{x}$. Therefore, it must be the case that $\displaystyle\int_1^\infty \frac{x^4}{x^5-3x}\,dx > \int_1^\infty \frac{1}{x}\,dx$. We then evaluate this new integral:

$$\int_1^\infty \frac{1}{x}\,dx = \lim_{t\to\infty}\left[\int_1^t \frac{1}{x}\,dx\right] = \lim_{t\to\infty}\left[\left(\ln|x|\right)\Big|_1^t\right] = \lim_{t\to\infty}\left[\ln(t) - \ln(1)\right] = \lim_{t\to\infty}\left[\ln(t)\right] = \infty$$

Since the integral $\displaystyle\int_1^\infty \frac{1}{x}\,dx$ diverges, then, by the Comparison Theorem, the integral $\displaystyle\int_1^\infty \frac{x^4}{x^5-3x}\,dx$ must also diverge.

b) $\displaystyle\int_1^\infty \frac{x^3+x+1}{x^5+2x+1}\,dx$

For $x \geq 1$ we see that $\dfrac{x^3+x+1}{x^5+2x+1} < \dfrac{x^3+x+1}{x^5} = \dfrac{1}{x^2} + \dfrac{1}{x^4} + \dfrac{1}{x^5}$. Therefore it must be the case that $\displaystyle\int_1^\infty \frac{x^3+x+1}{x^5+2x+1}\,dx < \int_1^\infty \left(\frac{1}{x^2} + \frac{1}{x^4} + \frac{1}{x^5}\right)dx$. We then evaluate this new integral:

$$\int_1^\infty \left(\frac{1}{x^2} + \frac{1}{x^4} + \frac{1}{x^5}\right)dx = \lim_{t\to\infty}\left[\int_1^t \left(\frac{1}{x^2} + \frac{1}{x^4} + \frac{1}{x^5}\right)dx\right] = \lim_{t\to\infty}\left[\left(-\frac{1}{x} - \frac{1}{3x^3} - \frac{1}{4x^4}\right)\Big|_1^t\right]$$

$$= \lim_{t\to\infty}\left[\left(-\frac{1}{t} - \frac{1}{3t^3} - \frac{1}{4t^4}\right) - \left(-\frac{1}{1} - \frac{1}{3} - \frac{1}{4}\right)\right] = 0 + 1 + \frac{1}{3} + \frac{1}{4} = \frac{19}{12}$$

Since the integral $\displaystyle\int_1^\infty \left(\frac{1}{x^2} + \frac{1}{x^4} + \frac{1}{x^5}\right)dx$ converges, then, by the Comparison Theorem, the integral $\displaystyle\int_1^\infty \frac{x^3+x+1}{x^5+2x+1}\,dx$ must also converge.

c) $\displaystyle\int_1^\infty \frac{x+1}{\sqrt{x^4-x}}\,dx$

For $x \geq 1$ we see that $\dfrac{x+1}{\sqrt{x^4-x}} > \dfrac{x+1}{\sqrt{x^4}} > \dfrac{x}{\sqrt{x^4}} = \dfrac{x}{x^2} = \dfrac{1}{x}$. Therefore, it must be the case that

$\displaystyle\int_1^\infty \frac{x+1}{\sqrt{x^4-x}}\,dx > \int_1^\infty \frac{1}{x}\,dx$. We then evaluate this new integral:

$$\int_1^\infty \frac{1}{x}\,dx = \lim_{t\to\infty}\left[\int_1^t \frac{1}{x}\,dx\right] = \lim_{t\to\infty}\left[\ln|x|\Big|_1^t\right]$$

$$= \lim_{t\to\infty}\left[\ln(t) - \ln(1)\right] = \infty$$

Since the integral $\displaystyle\int_1^\infty \frac{1}{x}\,dx$ diverges, then, by the Comparison Theorem, the integral $\displaystyle\int_1^\infty \frac{x+1}{\sqrt{x^4-x}}\,dx$ must also diverge.

d) $\displaystyle\int_0^\pi \frac{\sin^2(x)}{\sqrt{x}}\,dx$

For $x > 0$ we see that $\dfrac{\sin^2(x)}{\sqrt{x}} \leq \dfrac{1}{\sqrt{x}}$. Therefore, it must be the case that $\displaystyle\int_0^\pi \frac{\sin^2(x)}{\sqrt{x}}\,dx \leq \int_0^\pi \frac{1}{\sqrt{x}}\,dx$. We then evaluate this new integral:

$$\int_0^\pi \frac{1}{\sqrt{x}}\,dx = \lim_{t\to 0^-}\left[\int_t^\pi \frac{1}{\sqrt{x}}\,dx\right] = \lim_{t\to 0^-}\left[2\sqrt{x}\Big|_t^\pi\right] = \lim_{t\to 0^-}\left[2\sqrt{\pi} - 2\sqrt{t}\right] = 2\sqrt{\pi}$$

Since the integral $\displaystyle\int_0^\pi \frac{1}{\sqrt{x}}\,dx$ converges, then, by the Comparison Theorem, the integral $\displaystyle\int_0^\pi \frac{\sin^2(x)}{\sqrt{x}}\,dx$ must also converge.

e) $\int_0^{\infty} \frac{x}{x^3+1}\,dx$

First we note that $\int_0^{\infty} \frac{x}{x^3+1}\,dx = \int_0^{1} \frac{x}{x^3+1}\,dx + \int_1^{\infty} \frac{x}{x^3+1}\,dx$.

(We split the integral here to avoid issues with the bound of 0 in the integral that we plan to compare with. This is a determination made after one would experiment with evaluating the comparison integral.)

For $x \geq 0$ we see that $\frac{x}{x^3+1} < \frac{x}{x^3} = \frac{1}{x^2}$. Therefore, it must be the case that $\int_1^{\infty} \frac{x}{x^3+1}\,dx < \int_1^{\infty} \frac{1}{x^2}\,dx$. We then evaluate this new integral:

$$\int_1^{\infty} \frac{1}{x^2}\,dx = \lim_{t\to\infty}\left[\int_1^{t} \frac{1}{x^2}\,dx\right] = \lim_{t\to\infty}\left[-\frac{1}{x}\bigg|_1^t\right] = \lim_{t\to\infty}\left[-\frac{1}{t}+1\right] = 1$$

Since the integral $\int_1^{\infty} \frac{1}{x^2}\,dx$ converges, then, by the Comparison Theorem, the integral $\int_1^{\infty} \frac{x}{x^3+1}\,dx$ must also converge. Since $\int_0^{1} \frac{x}{x^3+1}\,dx$ clearly converges, we have that $\int_0^{\infty} \frac{x}{x^3+1}\,dx$ converges.